

OTA-C based

The Second Order low pass filter

its transfer function is:

$$H_{LP}(s) = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

s = Laplace variable

ω_0 = Angular frequency

Q = Quality factor

OTA equation:-

ideal OTA produces output current:

$$i_o = g_m (V_+ - V_-)$$

g_m = transconductance

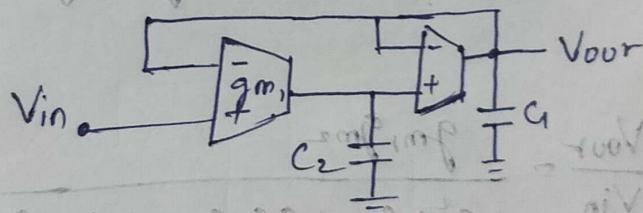
Capacitor equation:-

$$i_c = C \frac{dv}{dt}$$

i_c = current through capacitor

$$i_c = sCV$$

OTA + capacitor can behave like integrator



The equation of C_2 :

$$i_2 = g_{m2} (V_{in} - V_{out})$$

$$i_{C2} = sC_2 V_x$$

$$\therefore sC_2 V_x = g_{m2} (V_{in} - V_{out}) \rightarrow (1)$$

The equation of C_1 :

$$i_1 = g_{m1} (V_x - V_{out})$$

$$sC_1 V_{out} = g_{m1} (V_x - V_{out})$$

$$sC_1 V_{out} = g_{m1} (V_x - V_{out}) \rightarrow (2)$$

$$sC_1 V_{out} = g_{m1} V_x - g_{m1} V_{out}$$

$$sC_1 V_{out} + g_{m1} V_{out} = g_{m1} V_x$$

$$V_{out} (sC_1 + g_{m1}) = g_{m1} V_x$$

$$V_x = \left(\frac{sC_1 + g_{m1}}{g_{m1}} \right) V_{out} \rightarrow (3)$$

Substitute @ 3 in 1?

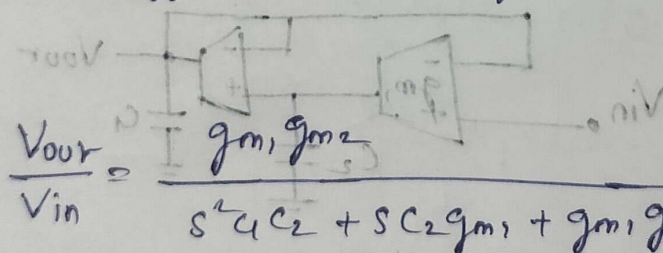
$$sC_2 V_x = g_{m2} (V_{in} - V_{out})$$

$$sC_2 \left(\frac{sC_1 + g_{m1}}{g_{m1}} \right) V_{out} = g_{m2} (V_{in} - V_{out})$$

$$sC_2 (sC_1 + g_{m1}) V_{out} = g_{m1} g_{m2} (V_{in} - V_{out})$$

$$(s^2 C_1 C_2 + sC_2 g_{m1}) V_{out} = g_{m1} g_{m2} V_{in} - g_{m1} g_{m2} V_{out}$$

$$V_{out} (s^2 C_1 C_2 + sC_2 g_{m1} + g_{m1} g_{m2}) = g_{m1} g_{m2} V_{in}$$



$$\frac{V_{out}}{V_{in}} = \frac{g_{m1} g_{m2}}{s^2 C_1 C_2 + sC_2 g_{m1} + g_{m1} g_{m2}}$$

Divide with $C_1 C_2$ with everything

$$\frac{V_{out}}{V_{in}} = \frac{\frac{g_{m1} g_{m2}}{C_1 C_2}}{s^2 + s \frac{g_{m1}}{C_1} + \frac{g_{m1} g_{m2}}{C_1 C_2}}$$

$$\therefore H(s) = \frac{g_{m1} g_{m2}}{C_1 C_2} \frac{1}{s^2 + \frac{g_{m1}}{C_1} s + \frac{g_{m1} g_{m2}}{C_1 C_2}} \quad \text{--- (4)}$$

Compare (4) with second order equation

$$\omega_0^2 = \frac{g_{m1} g_{m2}}{C_1 C_2}$$

$$\omega_0 = \sqrt{\frac{g_{m1} g_{m2}}{C_1 C_2}}$$

$$\frac{\omega_0}{Q} = \frac{g_{m1}}{C_1}$$

$$Q = \frac{\omega_0 C_1}{g_{m1}}$$

$$Q = \frac{C_1}{g_{m1}} \sqrt{\frac{g_{m1} g_{m2}}{C_1 C_2}}$$

$$Q = \sqrt{\frac{g_{m2} C_1}{g_{m1} C_2}}$$